

CALLIANNA KAYLIN PRIEBE

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EDUCATION

University of Colorado Boulder

Spring 2026

BA Computer Science | Philosophy Minor

GPA: 3.781

Relevant Coursework: Machine Learning, Artificial Intelligence, AI Ethics, CyberSecurity, Database Systems, Cognitive Science, History and Philosophy of Physics, Foundations of Quantum Engineering

INTERESTS & FOCUS

Interested in research and development for Artificial Intelligence and applied machine learning, with recent experience in applied computer vision, quantum computing, and data analysis for medical imaging.

Adaptive to complex processes, systems, and collaborative work environments. Seeking opportunities in Artificial Intelligence research and application.

Skills	Tools / Technologies	* = proficient
Machine & Deep Learning	Python* (NumPy*, Pandas*, TensorFlow*, PyTorch*, Graspologic), Scikit*, YOLO*, Hugging Face Transformers*, RAG	
Data Visualization	Matplotlib*, Seaborn, Plotly	
Software Engineering	C/C++*, C#, Java, SQL* (PostgreSQL, NoSQL, MySQL), Scala, Flask*, Swift, PHP, JSON, XML, Linux, Shell Scripting*, TypeScript, Docker, AWS	
Hardware Prototyping	Arduino*, Raspberry Pi, Oculus, Laptop assembly	
Dev Tools	Git*, Jupyter*, Colab*, Visual Studio*, PyCharm, Google Suite*, Postman, Unity, AutoCAD, ApplInventor, Microsoft Office*	
AI Agents	Claude Code*, Cursor*, Gemini*	
Quantum Computing	Qiskit, QuTiP	
UX & Front-End Development	HTML5/CSS* (Bootstrap*, Tailwind), JavaScript* (jQuery, Node, React*, Angular, Next)	

SELECTED EXPERIENCE

Applied Machine Learning Teaching Assistant - University of Colorado Boulder | Boulder, CO | Jan 2026 – May 2026

- Independently developed ML curriculum and taught this material in lectures to third-and-fourth year college students in CU Boulder's Applied Machine Learning course.
- Selected to work alongside ML Professor Featherstone to help correct and update course curriculum, grade material, and meet with students independently to privately tutor.
- Curriculum focused on Python ML (Scikit, TensorFlow, YOLO, HF Transformers) and data visualization.

Lead AI/ML Developer – Arterial (TechStars-backed Startup) | Boulder, CO | Feb 2025 – Jul 2025

- Won first place in TechStars Startup Weekend AI & Life Sciences Winston-Salem Feb-Mar 2025.
- Led the development and training of computer vision models for road hazard detection, achieving a mAP of 0.65, with a focus on classifying and prioritizing surface deformities. Focus in TypeScript & Python.
- Designed preprocessing pipelines and configured datasets for model optimization.

Machine Learning Intern – Microsoft Research | San Diego, CA (Remote) | April 2022 – Aug 2022

- Supported research using unsupervised ML to analyze MEG/EEG brain scans, developing anomaly detection models for early stroke and epilepsy spike prediction, processing over 300-channel signals sampled at 1 kHz.
- Built Python-based preprocessing pipelines for signal filtering (0.5-50 Hz), downsampling, and statistical analysis.
- Constructed graph-based neural representations of MEG data by converting time-sliced correlations into temporal graphs for application of MASE and Mahalanobis-based anomaly detection methods.

Software Developer – Hands-On Technology Education | San Diego, CA | 2015 – 2022

- At 17 y.o., developed curriculum for STEM courses for San Diego Unified School District, impacting thousands of students. Funded by San Diego County Office of Education.
 - Served as STEM instructor and mentor, teaching engineering fundamentals (software and hardware) to students ages 8–13, including Python, Arduino, AI, VR, and App design.
 - Independently engineered industry-level mechanics in AI, VR, and Python fundamentals into teachable material for middle and high school aged students.
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SELECTED PROJECTS

AI Mancala Agent – Machine Learning Course Project

Developed a reinforcement learning agent in Python capable of learning optimal strategies for Mancala. Utilized TensorFlow for model training and heuristic evaluation.

Quantum Shor’s Algorithm Simulation – UC Irvine, California State Summer School for Mathematics and Science (COSMOS): Computation and Machine Learning: Physics, Big Data, and Our Understanding of the Universe

In a team of 4, built a quantum circuit on a UCI virtual machine simulating Shor’s algorithm to compare classical vs. quantum computational efficiency. Successfully modeled Shor’s algorithm, given the processing limitations of running on classical hardware.

“GreenCycle” Recycling App – Cal State San Marcos Women’s Hackathon (Grand Prize Winner)

Developed a full-stack prototype promoting community recycling through gamified incentives. Won Grand Prize (youngest participant by 2 years). Had to get special permission to enter the competition due to age. (Nov. 2016, 12 y.o.)

CERTIFICATIONS & AWARDS

- First Place at TechStars Startup, Arterial AI Project
- Front-End Web Development Certificate, UCSD Extension (12-month course)
- Grand Prize, Cal State San Marcos Women’s Hackathon (2016)